Rethinking Decoder Redundancy in Efficient 3D Medical Image Segmentation: A Systematic Characterization Study

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Abstract

The decoder is an essential module for fusing multi-scale features and restoring spatial detail in U-Net architectures for 3D medical image segmentation. Recent designs, especially attention-based decoders, sharply increase its cost, and efficient methods such as EffiDec3D respond by shrinking high-resolution decoder stages—removing $> 90\%$ *of its parameters and FLOPs across 3D UX-Net, MedNeXt-M-K3 and SwinUNETR with little loss in Dice. This success, however, rests on the empirical assumption that decoder computation is static and globally redundant; it does not explain* why *the computation is removable,* where *the decoder still matters, or whether those regions can be identified at test time. We present a systematic characterization of decoder redundancy along three axes*—heterogeneity, concentration, *and* predictability—*by counting per-voxel prediction* flips *between matched full and efficient decoders, with a frozen shared-encoder control that attributes the changes to the decoder alone and a channel/resolution factor decomposition that separates the two sources of redundancy. We find the full decoder's net benefit is small and bidirectional (about* 0.1% *of voxels), spatially concentrated at organ boundaries and small structures, and identifiable from the efficient model's own uncertainty—entropy-ranked regions recover its net benefit at roughly five times the matched-random rate at a fixed budget. These trends are consistent across three architectures and three datasets, indicating that decoder redundancy is structured and predictable rather than uniform. Our analysis provides an evaluation protocol and design guidance for more efficient, region-adaptive segmentation decoders.*

1. Experimental Setup

Experiment plan. Our evaluation study reproduces EffiDec3D [11] and evaluates it with an *L-shaped* protocol (Table 1) that separately tests cross-dataset and cross-architecture consistency without an expensive full factorial. A **dataset axis** runs the matched 3D UX-Net [8] pair on three diverse tasks from the EffiDec3D evaluation—BTCV abdominal CT, FeTA [10] fetal-brain MRI, and MSD Task08 HepaticVessel [1] (CT, thin vessels/small lesions)—and an **architecture axis** runs matched pairs on BTCV for three backbone families: 3D UX-Net (large-kernel CNN), Swin UNETR [13] (hierarchical Transformer), and MedNeXt-M-K3 [12] (ConvNeXt-style CNN). The two axes share the 3D UX-Net/BTCV anchor, which also carries the mechanism analysis below.

Table 1. L-shaped protocol. The 3D UX-Net row tests cross-dataset consistency; the BTCV column tests cross-architecture consistency; they meet at the UX-Net/BTCV anchor ($*$, completed). Off-axis cells (–) are not run.

Backbone	BTCV (CT)	FeTA (MRI)	HepaticVessel (CT)
3D UX-Net	✓*	✓	✓
Swin UNETR	✓	–	–
MedNeXt-M-K3	✓	–	–

Dataset and split. The primary task is the 13-organ BTCV/Synapse abdominal CT benchmark with 18 training and 12 validation volumes: spleen, right and left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal and splenic veins, pancreas, and right and left adrenal glands. Images are intensity-windowed and resampled following the public EffiDec3D pipeline; the same validation subjects are paired across all model comparisons on a given dataset.

Two comparison regimes. Within each cell the full and EffiDec3D models share the same encoder *architecture* and differ in the decoder (channels reduced to the minimum encoder width, additive skip aggregation, resolution factor two). We analyze them under two regimes. (i) **End-to-end (ecological).** Both models are trained independently, so they share the encoder architecture but not its weights; their flips reflect the deployed full-vs-efficient difference but conflate decoder capacity with encoder-weight and optimization differences. (ii) **Frozen shared encoder (causal).** We take one trained encoder, freeze it, and train the decoder variants on *identical* features, so prediction differences are attributable to the decoder alone. Both regimes hold the encoder architecture fixed, so the study characterizes the decoder and draws no conclusions about the encoder itself.

Frozen-encoder factor decomposition. EffiDec3D removes decoder computation in two ways at once—narrower channels and a dropped high-resolution stage. To attribute effects, we decompose them with a 2×2 factorial on the frozen encoder, all within the EffiDec3D decoder family so only the two knobs vary: full (highest-resolution stage, wide channels), channel-only (wide $\to$ 48 channels), resolution-only (drop the high-resolution stage), and the combined EffiDec3D decoder. Pairwise flips (Full→channel-only, Full→resolution-only, Full→combined) isolate which factor changes which regions. The frozen encoder is optimized for the full decoder, so "efficient still matches on frozen features" is a *conservative* redundancy result.

Null-pair seed control. Because the net flip is tiny, we bound seed noise with a same-architecture null pair: two Full models trained with different (predeclared) seeds. The claim is that Full-vs-EffiDec3D net flips and their concentration exceed this Full-vs-Full null.

Implementation. We reproduce the EffiDec3D training protocol on a single NVIDIA RTX 5090 (32 GB, Blackwell) under PyTorch 2.6 / CUDA 12.8 with MONAI, using BF16 mixed precision (no loss scaling). Each model is trained for 45,000 optimizer steps (validation every 250) with AdamW at learning rate 10^{-3}, a Dice–cross-entropy loss, four 96^3 random-crop samples per volume, and full in-memory caching. Sliding-window inference uses a 96^3 window, batch size four, overlap 0.7, and constant blending. We evaluate one full run and two independently initialized EffiDec3D runs per cell to expose seed sensitivity. Segmentation quality is reported with Dice and 95th-percentile Hausdorff distance (HD95), and efficiency with parameters, multiply–accumulate operations (MACs), latency, and peak memory.

Core quantities. Let $\hat{y}_{\mathrm{E}}(v)$, $\hat{y}_{\mathrm{F}}(v)$, and $y(v)$ be the efficient prediction, full prediction, and ground-truth label at voxel v. We measure where the full decoder changes predictions with *prediction flips*, the standard tool for comparing two classifiers on paired data [2, 14]:

$$P(v) = \mathbb{1}[\hat{y}_{\mathrm{E}}(v) \neq y(v) \ \wedge \ \hat{y}_{\mathrm{F}}(v) = y(v)], \quad (1)$$

$$N(v) = \mathbb{1}[\hat{y}_{\mathrm{E}}(v) = y(v) \ \wedge \ \hat{y}_{\mathrm{F}}(v) \neq y(v)], \quad (2)$$

$$U(v) = P(v) - N(v), \quad (3)$$

a *positive* flip (full corrects efficient), a *negative* flip (full degrades efficient), and the *net* flip. P and N are exactly the discordant cells of the paired-agreement table behind McNemar's classifier-comparison test [3, 9]; volume rates are $R_{\mathrm{pos}} = \mathrm{mean}_v P(v)$, $R_{\mathrm{neg}} = \mathrm{mean}_v N(v)$, and $R_{\mathrm{net}} = R_{\mathrm{pos}} - R_{\mathrm{neg}}$. As test-time signals computable from the efficient model alone we use predictive entropy $H(v) = -\sum_c p_c(v) \log p_c(v)$, confidence $\max_c p_c(v)$, and Monte-Carlo dropout variance [4].

Characterization metrics. We group the analysis into three questions; each metric is computed per grid cell and aggregated over subjects with a bootstrap, since the twelve validation volumes—not the correlated voxels—are the independent units.

(1) Heterogeneity: is the benefit evenly distributed?

- **O1, boundary error ratio.** Mean voxel error on the eroded boundary band divided by the eroded-interior error; >1 indicates boundary-concentrated error.
- **O4, per-organ difficulty.** Per-class Dice and mean entropy H, ranking the structures the decoder struggles on.
- **O10, size–difficulty relation.** Spearman ρ between organ volume and difficulty, and the partial correlation of H and error given $\log$-volume.
- **Boundary-resolved flips.** Positive/negative/net flip rate binned by Euclidean distance to the nearest GT boundary—testing whether the decoder *benefit* (not just error) concentrates near boundaries. Complemented by Normalized Surface Dice (NSD) at 1–2 voxel tolerance for a surface-level view.

(2) Concentration: does a small subset carry most net benefit?

- **O2, entropy sparsity.** $\Pr[H(v) > \tau]$, the fraction of high-entropy voxels.
- **O5, uncertainty–benefit correlation.** Subject-level Pearson $r(H, U)$ between binned entropy and net flips.
- **O9$^{\text{orc}}$, voxel recovery.** $\sum_{\text{top-}k\%} P(v) \,/\, \sum P(v)$, the share of positive flips in the highest-entropy $k\%$ of voxels—a non-deployable upper bound.
- **O6, training persistence.** Mean entropy across checkpoints, testing whether concentration is a training transient.

(3) Predictability: can a test-time signal find the regions?

- **O3, flip discrimination and calibration.** AUROC and AUPRC of entropy predicting flips [7], and expected calibration error (ECE) [6].

Table 2. Reproduction fidelity of the matched 3D UX-Net pair, per dataset. We report mean Dice of our runs against the published EffiDec3D value; Δ is our Effi-UX minus the published Effi-UX. Data were reconstructed with identity affine metadata, so this is a controlled reimplementation, not an exact reproduction.

Dataset	Full-UX	Effi-UX	pub. Effi-UX	Δ
BTCV (CT, 13-organ)	.792	.773	.793	−.020
FeTA (MRI)	–	–	–	–
HepaticVessel (CT)	–	–	–	–

[TODO: Fill FeTA/HepaticVessel rows as the dataset-axis cells complete.]

- **O9, opportunity curve.** A risk–coverage curve [5] over contiguous 16^3 blocks (plus a 4-voxel halo) ranked by entropy, reporting net-flip utility recovered versus a matched-random selector.
- **O11, routing-signal comparison.** Subject-level correlation of entropy, confidence, and MC-dropout with net flips.

Cross-dataset (**O7**) and cross-backbone (**O8**) consistency re-evaluate these metrics across the rows and columns of Table 1.

[TODO: Completed results below are the 3D UX-Net/BTCV grid cell. Remaining cells (other backbones and datasets) are in progress; cross-backbone (columns) and cross-dataset (rows) consistency are reported once enough cells are filled.]

2. Results

2.1. Controlled Reimplementation

We reproduce the matched Full-UX/Effi-UX pair and report its fidelity per dataset (Table 2). On BTCV our Effi-UX reaches 0.773 mean Dice over two seeds, 1.7–2.2 points below the published 0.793, while Full-UX is within 0.6 points of its published value; the residual gap reflects our reconstructed identity-affine data rather than the paper's processed set, and a FP32-versus-BF16 check ruled out a numerical cause. Efficiency reproduces the published reductions: Effi-UX uses 14.1× fewer MACs, 17.9× fewer parameters, $\sim 5\times$ lower latency, and $\sim 7.7\times$ lower peak memory. The same pattern holds on the architecture axis (Table 3): on BTCV, Swin UNETR reproduces its published full model (0.805 vs. 0.8013) and its EffiDec3D decoder cuts MACs 6.5× at a 2.6-point Dice cost—a smaller reduction than 3D UX-Net's, matching EffiDec3D's own per-backbone figures. MedNeXt-M-K3, the strongest backbone at 0.837 full Dice, shows the same 2.3-point cost at a 2.2× MAC reduction; across all three backbones the Dice cost is tightly consistent (−2.2/−2.6/−2.3%) even as the achievable MAC reduction ranges from 2.2× to 14.1×—consistent across architectures, not universal in magnitude.

Table 3. Architecture axis (BTCV): reproduction and efficiency of the matched full/EffiDec3D pair per backbone. Dice is Full→EffiDec3D; reductions are Full/EffiDec3D.

Backbone	Dice (F→E)	MAC↓	Param↓	Lat↓
3D UX-Net	.792→.773	14.1×	17.9×	5.4×
Swin UNETR	.805→.779	6.5×	5.7×	2.3×
MedNeXt-M-K3	.837→.815	2.2×	3.0×	2.2×

Table 4. Prediction-flip rates across efficient-decoder seeds (fractions of evaluated voxels). Positive/negative flips are the discordant cells of the paired comparison [9, 14].

Metric	seed 0	seed 1
Positive flip rate R_{pos}	.00339	.00290
Negative flip rate R_{neg}	.00209	.00225
Net flip rate R_{net}	.00130	.00065
Net flip rate, subject 95% CI	–	[.00017, .00113]

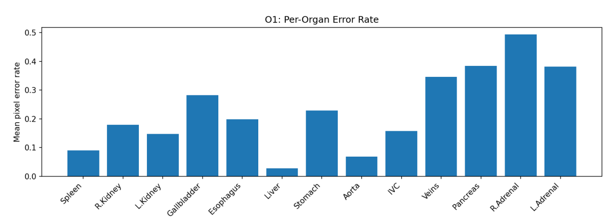


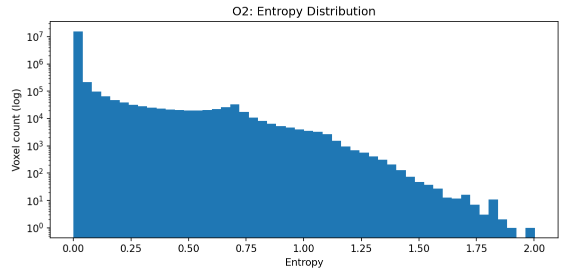
Figure 1. Per-organ error rate for Effi-UX (seed 1). Difficulty is strongly uneven across anatomy.

Treating these matched models as fixed, the rest of the paper characterizes *where* the full decoder's extra computation changes predictions—the question EffiDec3D's aggregate evaluation leaves open.

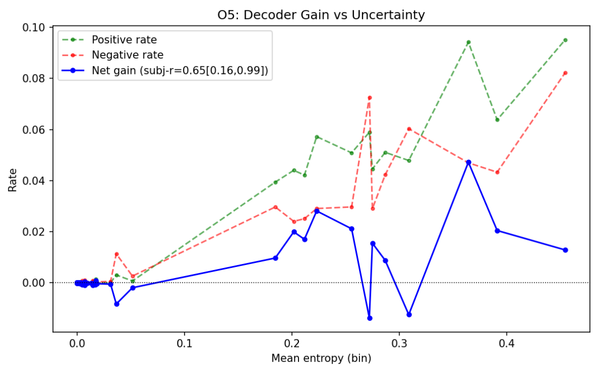
2.2. Heterogeneity Metrics

The full decoder's benefit is small and *bidirectional*: Full-UX corrects 0.290% of Effi-UX's voxels (R_{pos}) while degrading 0.225% (R_{neg}) for seed 1, and 0.339%/0.209% for seed 0, so the net flip rate R_{net} is only 0.065–0.130% of voxels (Table 4). Reporting positive flips alone would overstate the decoder's benefit roughly twofold.

This small net benefit is far from uniformly distributed. Error concentrates at structure boundaries: mean boundary error is 0.1892 versus 0.0481 in the eroded interior, a 3.93× pooled ratio (subject-mean 8.9×, 95% CI [5.8, 12.1]). It is also organ-dependent (Fig. 1), with the adrenal glands, pancreas, and veins hardest, and it scales with structure size: the organ-size/difficulty Spearman correlation is $\rho = -0.54$, and entropy still explains organ difficulty after controlling for log volume (partial $r = 0.81$). Decoder benefit is therefore heterogeneous across region, organ, and scale.



(a) Voxel entropy (log-count).



(b) Net flips vs. entropy.

Figure 2. Concentration of difficulty and net flips (seed 1). Uncertainty is spatially sparse (a), and net flips rise with it (b).

Crucially, resolving flips by distance to the GT boundary shows the *net flip* rate itself—not just error—peaks in the boundary *zone* rather than exactly at the surface: it is near zero at 0–1 vox (-0.0006), rises to its maximum of 0.034 at 1–2 vox, and decays to 0.006–0.008 beyond 4 vox; surface Dice (NSD) is $0.745/0.842$ at τ=1/2 mm. This concentration should exceed a same-architecture null pair (Full seed 0 vs. seed 1) [TODO: null-pair floor], ruling out seed noise.

2.3. Concentration Metrics

A small, high-uncertainty subset carries most of the net benefit. Predictive entropy is spatially sparse: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5 (Fig. 2a). Net flips concentrate in exactly this tail—the subject-level Pearson correlation between binned entropy and net flips is $r = 0.646$ (95% CI $[0.158, 0.994]$) for seed 1 and $r = 0.665$ ($[0.171, 0.996]$) for seed 0, both bounded above zero (Fig. 2b). A voxel-level oracle makes the concentration explicit: ranking foreground voxels by entropy, the top 20% contain 86.4% of all positive flips in both seeds. This concentration is not a transient of early training: mean entropy falls from 0.0279 at 5k iterations to 0.0104 at 45k, yet the sparse difficult residual persists.

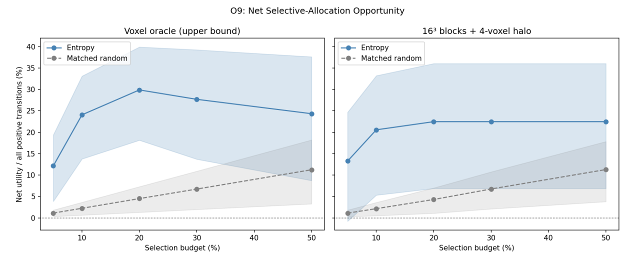


Figure 3. Region-level opportunity (seed 1). Contiguous 16^3 blocks ranked by entropy recover far more *net* flip utility than a matched-random selector at every budget $\geq 10\%$; the left panel is a non-deployable voxel oracle.

2.4. Predictability Metrics

Concentration is only actionable if the beneficial regions can be found without the ground truth. We assess this as an error-discrimination problem: predictive entropy should separate flip voxels from the rest [7]. Efficient-decoder entropy predicts the *positive-flip* voxel with per-subject AUROC 0.919 (95% CI $[0.898, 0.936]$) and AUPRC 0.312 ($[0.269, 0.356]$); predicting the error voxel—a cheaper baseline—gives AUROC 0.892 ($[0.863, 0.914]$) and AUPRC 0.578 ($[0.545, 0.613]$), and entropy is moderately calibrated (ECE 0.097, $[0.074, 0.124]$) [6]. The pooled-bin entropy–error correlation $r = 0.97$ is retained only as a descriptive diagnostic.

We then convert this into a deployable, selective-prediction analysis [5]. Non-overlapping 16^3 blocks are ranked by mean entropy and expanded by a 4-voxel halo; we measure net flip utility recovered against a matched-random selector under a paired subject bootstrap (Fig. 3). At the predeclared 20% block budget—executing 29.1% of the foreground-union volume after halo expansion—entropy-ranked regions attain a net utility of 0.225 versus 0.043 for random, a roughly fivefold gain, with a paired 95% lower bound of $0.056 > 0$; the advantage holds at every budget $\geq 10\%$ and vanishes only at 5%. These regions capture all positive flips but also 77.5% of the negative ones, so the *net* gain, not positive recovery, is the honest headline.

Finally, we compare cheap routing signals against held-out net flips per subject. Entropy and confidence track net flips almost identically—subject-level correlations 0.646 ($[0.158, 0.994]$) and 0.651 ($[0.153, 0.997]$), a paired difference of at most 0.01 ($[-0.009, -0.002]$)—so they are interchangeable for routing, and Monte-Carlo dropout is invalid for this checkpoint because the architecture has no active dropout (its stochastic variance is effectively zero). We therefore retain confidence as a required baseline and do not claim entropy is uniquely optimal.

2.5. Mechanism: Which Factor Is Removable

The comparisons above use end-to-end models, which share the encoder architecture but not its weights. As a further, causally cleaner check, we *freeze* one trained encoder and train the 2×2 decoder factorial on identical features, then compare Full→channel-only, Full→resolution-only, and Full→combined EffiDec3D [TODO: report per-factor flip rates and boundary profiles from the frozen-encoder run; hypothesis to test: dropping the high-resolution stage yields flips concentrated nearer the boundary, while channel reduction is spatially diffuse and smaller, which would explain why both are removable with little aggregate Dice change]. Because the frozen encoder was optimized for the full decoder, a near-match by the efficient decoder on those features would be a *conservative* statement of redundancy. We present this regime as validation of the decoder attribution rather than as the primary result.

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